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Question 1 [10 marks]

Design the **MarvelHero** class that extends the given **Hero** class with additional attributes and methods so that the driver code can generate the output below.

N.B. Heroes can have a power limit of 1200

```
#Parent Class
public class Hero {
    private String name;
    private int powerLevel;
    public static int heroCount = 0;

    public Hero(String name, int powerLevel) {
        this.name = name;
        this.powerLevel = powerLevel;
    }

    public String toString() {
        return "Hero Name: " + name + "\nPower level: " + powerLevel;
    }
}
```

```
#DriverCode

public class TestMarvelHero {
    public static void main(String[] args) {
        MarvelHero ironMan = new MarvelHero("Iron
Man", 500, "Tech");
        System.out.println(ironMan);
        System.out.println("-----1-----");
        MarvelHero thor = new MarvelHero("Thor", 800,
"God");
        MarvelHero hulk = new MarvelHero("Hulk", 700);
        System.out.println(hulk);
        System.out.println("-----2-----");

        System.out.println(MarvelHero.registeredHeroes());
        System.out.println("-----3-----");
        MarvelHero.battle(ironMan, thor);
        System.out.println("-----4-----");
        MarvelHero.battle(hulk, ironMan);
        System.out.println("-----5-----");
        ironMan.upgradeWeapon("Bleeding Edge", 300);
        System.out.println(ironMan);
        System.out.println("-----6-----");
        thor.upgradeWeapon("Stormbreaker", 400);
        System.out.println(thor);
        System.out.println("-----7-----");
    }
}
```

```
#Output
Hero Name: Iron Man
Power level: 500
Origin: Tech
Current Gear: Basic
-----1-----
Hero Name: Hulk
Power level: 700
Origin: Unknown
Current Gear: Basic
-----2-----
Total Heroes: 3
All registered hero details:
Iron Man,Thor,Hulk,
-----3-----
Thor is victorious.
Thor's power level is greater.
-----4-----
Hulk is victorious.
Hulk's power level is greater.
-----5-----
Hero Name: Iron Man
Power level: 800
Origin: Tech
Current Gear: Bleeding Edge
-----6-----
```

```

        MarvelHero.battle(thor, ironMan);
        thor.upgradeWeapon("Mjolnir", 500);
        System.out.println("-----8-----");
        MarvelHero spiderMan = new
MarvelHero("Spider-Man", 400, "Enhanced Human");
        MarvelHero blackWidow = new MarvelHero("Black
Widow", 350);
        MarvelHero doctorStrange = new
MarvelHero("Doctor Strange", 600, "Sorcerer");
        System.out.println("-----9-----");

System.out.println(MarvelHero.registeredHeroes());
    }
}

```

```

Hero Name: Thor
Power level: 1200
Origin: God
Current Gear: Stormbreaker
-----7-----
Thor is victorious.
Thor's power level is greater.
Cannot upgrade weapon beyond the
limit.
-----8-----
Hero was not registered.
Hero was not registered.
-----9-----
Total Heroes: 4
All registered hero details:
Iron Man,Thor,Hulk,Spider-Man,

```